Note on Algebraic Geometry YouTube Series

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§1 Disclaimer

This note is basically a copy of algebraic geometry note from Andreas Gathmann and algebraic geometry YouTube course by Seidon Alsaody, Uppsala University. This note is written for my own reference. So there may be some typos and mistakes. Please kindly point them out to me if you find any.

§2 Preliminaries

We assume our ground field k is algebraically closed throughout the note.
Preliminaries are basic abstract algebra. More specifically, you need to know

§2.1 Basic abstract algebra

- Rings
- Ideals: maximal, prime and radical ideals
- Fields and field extensions
- Noetherian Rings: three equivalent definitions and Hilbert's basis theorem.

Do not learn algebraic geometry if you are not comfortable with these concepts.

§2.2 Affine Space and Affine Algebraic Sets

- Affine Space $\mathbb{A}^n$: the set of all n-tuples of elements in k.
- Affine algebraic sets: $V(J)$ is the set of common zeros of all polynomials in J or the set of common intersection of curve defined by polynomials in J. A subset of $\mathbb{A}^n$ is algebraic if it is of the form $V(J)$ for some ideal J of $k[x_1, x_2, \ldots, x_n]$.
- Algebraic sets of $\mathbb{A}^1$ must be finite sets or the whole set because a nonzero polynomial in one variable has only finite many roots.
- Properties of $V(J)$:
  1. If $I \subseteq J$, then $V(J) \subseteq V(I)$
  2. $V(I) \cap V(J) = V(I + J)$. More generally, $\bigcap_\alpha V(I_\alpha) = V(\sum_\alpha I_\alpha)$ (all but finitely many I_α are zero).
  3. $V(I) \cup V(J) = V(IJ) = V(I \cap J)$ (Nontrivial, notice that $IJ \subseteq I \cap J$. They are not equal in general. Exercise: If $I + J = R$ then $IJ = I \cap J$.) (This generalizes to finite union but not infinite union. Counterexample: the union of all points but one in $\mathbb{A}^1$ is not algebraic set.)
  4. $V(\sqrt{J}) = V(J)$

§3 Hilbert's Nullstellensatz

Let $A = k[x_1, x_2, \ldots, x_n]$ be the polynomial ring over an algebraically closed field k.

- First, we need a reverse function of $V(J)$. Define $I(X)$ to be the ideal of A consisting of all polynomials that vanish on $X \subseteq \mathbb{A}_k^n$. That is all the polynomials f such that its curve passing through all points in X.

- Properties of $I(X)$:
  1. If $X \subseteq Y$, then $I(Y) \subseteq I(X)$
  2. $I(a_1, a_2, \ldots, a_n) = (x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$ (exercise, this is in fact not trivial)
  3. If X is an algebraic set, then $V(I(X)) = X$. (this is not hard)

- Nullstellensatz: For any ideal J of A, $I(V(J)) = \sqrt{J}$ Proof sketch:
  1. Zariski's Lemma: If F is a finitely generated k-algebra that is also a field, then F is a finite extension of k. There is an elementary proof if we assume k is uncountable (Key: consider $\frac{1}{x-\lambda}$ for $\lambda \in k$ must be linearly dependent because F is countable dimensional over k). We can use Noether normalization to prove this for general infinite k.
  2. Weak Version: $V(J) \neq \emptyset$ for all proper ideal J of A.
  3. Strong Version: Consider $J' = J + (1 - fy)$ in $k[x_1, \ldots, x_n, y]$.

- Some immediate consequences:
  1. $I(X \cup Y) = I(X) \cap I(Y)$
  2. $I(X \cap Y) = \sqrt{I(X) + I(Y)}$
  3. If $f = g$ on $\mathbb{A}^n$, then $f = g$ in A.
  4. If $f_{|X} = g_{|X}$, then $(f - g) \in I(X)$. We say $f \equiv g \mod I(X)$.

§4 Regular Function on Closed Set

Let $X \subseteq \mathbb{A}^n$ be an algebraic set.

§4.1 Coordinate Rings

- Definition: $\mathcal{O}(X) = \frac{k[x_1,x_2,\ldots,x_n]}{I(X)}$ is called the coordinate ring of X. This is clearly a k-algebra.

Example 4.1

$\mathcal{O}(\mathbb{A}^n) \cong A$
$\mathcal{O}(p) \cong k$ (evaluation at point p)
$\mathcal{O}(\ell) \cong k[t]$ where ℓ is a line in $\mathbb{A}^n$

- A function ϕ on X is called regular function or polynomial function if there exists $f \in A$ such that $\phi(x) = f(x)$ for all $x \in X$.
- There is a one-to-one correspondence between all the coordinate rings and all affine (affine means finitely generated) reduced k-algebras. Easy to show.

§4.2 Algebraic Subset of X

- Let J be an ideal of $\mathcal{O}(X)$. Define $V_X(J) = \{p \in X \mid f(p) = 0 \text{ for all } f \in J\}$. For any subset Y of X, Y is called an algebraic subset of X if there exists an ideal J of $\mathcal{O}(X)$ such that $Y = V_X(J)$.
- Let π be the projection from A to $\mathcal{O}(X)$. Notice that $V_X(J)$ is just $V(\pi^{-1}(J))$. So an algebraic subset of X is also an algebraic set in $\mathbb{A}^n$.
- We can also define $I_X(Y) = \{f \in \mathcal{O}(X) \mid f(p) = 0 \text{ for all } p \in Y\}$. We can easily verify $I_X(Y) = \frac{I(Y)}{I(X)}$.
- With this, we can easily verify the following properties:
  1. $\mathcal{O}(Y) \cong \frac{\mathcal{O}(X)}{I_X(Y)}$
  2. We have the relative version of Nullstellensatz: For any ideal J of $\mathcal{O}(X)$, $I_X(V_X(J)) = \sqrt{J}$. (The proof is just applying the original Nullstellensatz and playing around with the quotient of ideals: $\frac{\sqrt{J}}{I(X)} = \sqrt{\frac{J}{I(X)}}$. More generally, the preimage of a radical ideal is radical) The weak one is also true: the maximal ideals of $\mathcal{O}(X)$ are in one-to-one correspondence the maximal ideals of A that contains $I(X)$, which are in one-to-one correspondence with points in X.

Remark 4.2. Later on, we will just use I and V instead of I_X and V_X when there is no confusion (when the ground algebraic set X is fixed).

§5 Zariski's Topology

§5.1 Basic Definitions

- Definition: The closed sets of Zariski's topology on $\mathbb{A}^n$ are all the algebraic sets.

> **Remark 5.1.** The fact that algebraic sets form a topology is very surprising.

- If X is an algebraic set, we can define the Zariski's topology on X similarly: closed sets are all the algebraic subsets of X or the intersection of algebraic sets with X. (These two are the same)
- $\mathbb{A}^1$ is non-hausdorff, any nonempty open set is dense and has finite complement. In general, $\mathbb{A}^n$ is also non-hausdorff and any nonempty open set is dense (we will show it later).

§5.2 Connectedness

- An algebraic set X is connected if we cannot write $X = X_1 \cup X_2$ where X_1, x_2 are nonempty proper **disjoint** closed (closed under X or $\mathbb{A}^n$, doesn't matter since X itself is closed) subsets of X.
- Geometric intuition: connected means we can draw X without lifting the pen.
- Algebraic perspective: If $X = X_1 \cup X_2$ is disconnected, then $\mathcal{O}(X) \cong \mathcal{O}(X_1) \times \mathcal{O}(X_2)$ (this is just CRT)
- Some standard properties about connectedness:
  1. If a set X admits an open connected covering (U_i) such that $U_i \cap U_j \neq \emptyset$ for all i, j, then X is connected. (exercise)
  2. Connectedness is preserved under continuous maps.
  3. If X is connected, so is $\overline{X}$.

§5.3 Irreducibility

- An algebraic set X is irreducible if we cannot write $X = X_1 \cup X_2$ where X_1, x_2 are nonempty proper closed subsets of X.
- Irreducibility implies connectedness but not vice versa (counterexample: $V(xy)$ in $\mathbb{A}^2$ is clearly connected but irreducible $V(xy) = V(x) \cup V(y)$.)
- Geometric intuition: irreducible means we cannot separate X using "integral" several pieces.
- We have discussed that open sets of $\mathbb{A}^n$ are very large. For example, nonempty open set of $\mathbb{A}^1$ is always dense. In fact, we have the following lemma to formalize and generalize this idea.

Lemma 5.2

In any irreducible topological space X, any nonempty open set is dense in X.

Proof. Let U be a nonempty open set in X. Consider $\overline{U} \cap (X - U) = \emptyset$. Thus, $\overline{U} = X$. (This proof is purely topological.) □

> **Remark 5.3.** Let X be an irreducible algebraic set and U be a nonempty open set of X. If $f = g$ on U, then $f = g$ on X. This is because $V_X(f - g)$ is closed in X and contains U. Since $\overline{U} = X$, we have $X = \overline{U} \subseteq V_X(f - g) \subseteq X$. Thus, $f = g$ on X. This makes sense because open set is very large in irreducible algebraic set (dense).

- Algebraic perspective: X is irreducible if and only if $\mathcal{O}(X)$ is an integral domain if and only if $I(X)$ is a prime ideal of A.

  This is very nice! It allows us to study geometric irreducibility using algebraic tools: ideals and integral domains. With this characterization, we know all affine space $\mathbb{A}^n$ are irreducible just like $\mathbb{A}^1$.

- Some standard properties about irreducibility:
  1. Irreducibility is preserved under continuous maps.
  2. A set is irreducible if and only if its closure is irreducible.
  3. If a set X admits an open irreducible covering $(U_i)_{i \in I}$ such that $U_i \cap U_j \neq \emptyset$ for all i, j, then X is irreducible.
     Proof: Suppose $X = M \cap N$ where M and N are closed subsets of X, then consider $U_i = X \cap U_i = (M \cap U_i) \cup (N \cap U_i)$. Since U_i is irreducible, we conclude that U_i are either contained in M or N. Assume $J \subset I$ such that $U_i \subseteq M$ for all $i \in J$ and $U_i \subseteq N$ for all $i \notin J$. Consider $M' = M \cap (X - \bigcup_{i \notin J} U_i)$ and $N' = N \cap (X - \bigcup_{i \in J} U_i)$. We have $X = M' \cup N'$ and $M' \cap N' = \emptyset$. So X is disconnected unless one of $J = I$ or $J = \emptyset$. However, X must be connected as we have shown before. Thus, X is irreducible.
  4. If $X \subseteq \mathbb{A}^n$ is irreducible and $Y \subseteq \mathbb{A}^m$ is irreducible, then $X \times Y \subseteq \mathbb{A}^{n+m}$ is also irreducible.

§5.4 Noetherian Topological Space

- A topological space X is called Noetherian if every descending chain of closed subsets terminates.

Corollary 5.4

Any algebraic set $X \subseteq \mathbb{A}^n$ is a Noetherian topological space. This follows from the fact that $k[x_1, x_2, \ldots, x_n]$ is a Noetherian ring, and its quotient $\mathcal{O}(X)$ is also Noetherian.

There are six important properties of Noetherian topological spaces:

1. The subspace of a Noetherian topological space is also Noetherian. This is just like the submodule of a Noetherian module is also Noetherian.

2. Any Noetherian topological space can be decomposed into finite union of irreducible closed sets that are not contained in each other. This decomposition is unique up to permutation.

   With this fact, we can more or less assume our ground algebraic set X is irreducible because we can always decompose it into irreducible components.
3. Similarly, any Noetherian topological space can be decomposed into finite **disjoint** union of connected closed sets that are not contained in each other. This decomposition is unique up to permutation.
4. If X is the finite union of Noetherian topological spaces, then X is also Noetherian.
5. Noetherian is a topological property preserved under continuous maps.
6. **A topological space X is Noetherian if and only if every open subset of X is compact if and only if every subspace of X (induced topology from X) is compact.** This is tremendous because compactness is a very useful property in topology. Anyway, the proof is an exercise.

   This property makes sense because open sets are very large in Zariski's topology. So finite many open sets will be sufficient to cover an open set.

> **Remark 5.5.** Noether killed infinite. Big thank you to Noether.

§5.5 Regular Function on Zariski's Open Set

Given a zariski's topology on an algebraic set X,

- We have defined the regular function on algebraic sets/closed set of X: coordinate rings. Now, we wish to define it on open set. So we may assume X is an irreducible algebraic set, and we wish to define regular function on open set U of Zariski's topology of X.
  The problem is that U is dense in X, so intuitively, $I(U)$ would be 0 because U is so large. Indeed, $V_X(I_X(U)) = \overline{U} = X$ (this is true in general, $V(I(X)) = \overline{X}$). So $I_X(U) = I_X(X) = 0$. Alternative way to see this is that if f vanishes on U, then $U \subseteq \overline{U} \subseteq V_X(f) \subseteq X$. So f must be 0.
  Therefore, we can't define $\mathcal{O}_X(U)$ to be $\frac{\mathcal{O}(X)}{I_X(U)}$ because this is just $\mathcal{O}(X)$ again. Instead, we define it as follows:

- Definition: Let X be an algebraic set and U be a Zariski's open set of X. A function $f : U \to k$ is called a regular function on U if for each point $p \in U$, there exists an open neighborhood $p \in U_p \subseteq U$ and $g, h \in \mathcal{O}(X)$ such that for all $q \in U_p$, $f(x) = \frac{g(x)}{h(x)}$ and $h(x) \neq 0$ for all $x \in U_p$. The set of all regular functions on U is denoted by $\mathcal{O}(U)$. We can easily verify that $\mathcal{O}_X(U)$ (or just $\mathcal{O}(U)$ if the grounded algebraic set X is fixed) is a k-algebra.

> **Remark 5.6.** The notion of regular function on open set may lead to some confusion because X itself is an open set of X. So $\mathcal{O}(X)$ has two different meanings: the coordinate ring of X and the set of regular functions on the open set X. In fact,

> they are the same! We will see this in remark 4.11.

- We can also study the zero loci of a regular function on open set U.

Lemma 5.7
$\phi \in \mathcal{O}(U)$, then $V(\phi) = \{x \in U \mid \phi(x) = 0\}$ is closed in U.

Proof. For any $p \in U$, we can write $\phi = \frac{g_p}{h_p}$ on some open neighborhood U_p of p where $g_p, h_p \in \mathcal{O}(X)$ and h doesn't vanish on U_p. Consider $U_a \setminus V(\phi) = U_a \setminus V(g_p)$, which is open in U_a, thus open in U. So $U \setminus V(\phi) = \bigcup_{p \in U}(U_p \setminus V(\phi))$ is open in U. Therefore, $V(\phi)$ is closed in U. □

> **Remark 5.8.** A open set minus a closed set is still open. So this lemma is quite useful.

> **Remark 5.9** (Identity Theorem of regular functions)**.** With Lemma 4.6, we can see the following: If X is irreducible algebraic set and $U \subseteq V$ are nonempty open sets of X, then if $\phi_1, \phi_2 \in \mathcal{O}(V)$ and $\phi_{1|U} = \phi_{2|U}$, then $\phi_1 = \phi_2$ on V. This is becasue $V(\phi_1 - \phi_2)$ is closed in V and contains U. Since $\overline{V} = X$, V is irreducible. So $V = \overline{U} \subseteq V(\phi_1 - \phi_2) \subseteq V$. This makes sense because open set is very large in irreducible algebraic set (dense).

- The fact that the denominator depends on the choice of the point is quite annoying. It's a local definition. Here is an example:

Example 5.10
Let $X = V(x_1x_4 - x_2x_3) \subseteq \mathbb{A}^4$ and $U = X \setminus V(x_2, x_4)$. Then, we have the following regular function on U:

$$\phi = \begin{cases} \frac{x_1}{x_2} & \text{if } x_2 \neq 0 \\ \frac{x_3}{x_4} & \text{if } x_4 \neq 0 \end{cases}$$

This is clearly a regular function because it's locally a ratio of polynomials. But we can't write it globally as a ratio of polynomials. $k[x_1, x_2, x_3, x_4]/(x_1x_4 - x_2x_3)$ is non UFD.

§5.6 Regular function on principal open set

- However, we can restrict our open set to be principal open set to have a global property. First, we define principal open set:

Definition 5.11. Let X be an algebraic set and $f \in \mathcal{O}(X)$. Define $U_f = \{x \in X \mid f(x) \neq 0\}$. Then, U_f is called the principal open set of f in X.

There are two basic properties of principal open sets:

1. The intersection of two principal open sets is also a principal open set because $U_f \cap U_g = U_{fg}$.
2. Any open set U of X can be written as finite union of principal open sets. Trivial by Noetherian property.

- If U_f is a principal open set of $f \in \mathcal{O}(X)$, then we have a nice description of $\mathcal{O}(U_f) = \{\frac{g}{f^n} \mid g \in \mathcal{O}(X), n \in \mathbb{N}\}$. So we can write any regular function on U_f as a ratio of polynomials with denominator being some power of f, something global.

Proof. Consider a regular function $\phi : U_f \to k$. For any $p \in U_f$, we have open set U_p in U_f such that $p \in U_p \subseteq U_f$ and $\phi = \frac{g}{h}$ for some $g, h \in \mathcal{O}(X)$ and h doesn't vanish on U_p. We may further assume $U_p = U_t$ for some $t \in \mathcal{O}(X)$. The only thing we can do here is to multiply something on numerator and denominator to turn it into a better form. Consider $h' = h \cdot t$ and $g' = g \cdot t$. Then, we have $\phi = \frac{g'}{h'}$ on U_p. Notice that h' doesn't vanish on U_t but does vanish outside U_t. So $V(h') = V(t)$. So we may just assume $t = h'$. g' also vanishes outside U_t.

Since U_f is an open set in a Noetherian topological space, we have an finite open covering of U_f:

$$U_f = \bigcup_{i=1}^{m} U_{h_i}$$

Taking the complement we have:

$$V(f) = \bigcap_{i=1}^{m} V(h_i)$$

By Nullstellensatz, we have some $n \in \mathbb{N}$ such that $f^n \in (h_1, h_2, \ldots, h_m)$. So there exists $a_i \in \mathcal{O}(X)$ such that

$$f^n = \sum_{i=1}^{m} a_i h_i$$

Now, consider the following:

$$\psi = \frac{\sum a_i g_i'}{\sum a_i h_i}$$

For any $p \in U_f$, for any j, if $p \in U_{h_j}$, then $\frac{a_j g_j'}{a_j h_j}$ is defined at p and equals to $\phi(p)$. If $p \notin U_{h_j}$, then $h_j(p) = 0$, so $g_j'(p) = 0$ as well. So we can conclude $\psi(p) = \phi(p)$ (think about this step, look the first remark below). Therefore, $\psi = \phi$ on U_f. This shows that any regular function on U_f can be written as $\frac{g}{f^n}$ for some $g \in \mathcal{O}(X)$ and $n \in \mathbb{N}$.

□

> **Remark 5.12.** A key property about fraction is used here. If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$.

> **Remark 5.13.** Let $f = 1$. Then $U_f = X$. So we have $\mathcal{O}(X) = \{\frac{g}{1^n} \mid g \in \mathcal{O}(X), n \in \mathbb{N}\} = \mathcal{O}(X)$. This shows that the two different definitions of $\mathcal{O}(X)$ are indeed the same.

> **Remark 5.14.** This is clearly similar to localization of of $\mathcal{O}(X)$ at the multiplicative set generated by f. In fact, they are isomorphic. Proof is easy but the result is somewhere remarkable. It tells us that two regular functions agrees on a principal open set U_f if and only if they are equal as elements in the localization of $\mathcal{O}(X)$ at f. This is the ultimate identity theorem of regular functions on principal open set including algebraic sets.

- The regular functions on some non-principal open set may also be in a nice form. For example, we can consider $X = \mathbb{A}^2 \setminus \{0\}$ (nothing special about the origin here, we can remove any point). We claim that $\mathcal{O}(X) = k[x, y]$.

  Proof. Although X is not a principal open set, we can write it as union of two principal open sets: $X = U_x \cap U_y$.

  Any regular function on U_x can be written as $\frac{f(x,y)}{x^n}$ for some $f \in k[x, y]$ and $n \in \mathbb{N}$. Similarly, any regular function on U_y can be written as $\frac{g(x,y)}{y^m}$ for some $g \in k[x, y]$ and $m \in \mathbb{N}$. We assume $x \nmid f, g$ They must agree on $U_x \cap U_y = U_{xy}$, that is $f(x, y)y^m$ and $g(x, y)x^n$ agree on U_{xy} Since $\mathbb{A}^2$ is irreducible, by identity theorem of regular functions, we have

  $$f(x, y)y^m = g(x, y)x^n \text{ on } \mathbb{A}^2$$

  So

  $$f(x, y)y^m = g(x, y)x^n \text{ in } k[x, y]$$

  Since $k[x, y]$ is a UFD, we must have $x^n \mid f(x, y)$ and $y^m \mid g(x, y)$. Contradiction unless $n = m = 0$. Therefore, any regular function on X is actually a polynomial $f = g$ in $k[x, y]$. So $\mathcal{O}(X) = k[x, y]$. □

  > **Remark 5.15.** This is quite surprising because we have removed a point from $\mathbb{A}^2$ but the regular functions remain the same. This tells us that there is no rational function that is undefined only at one point in $\mathbb{A}^2$. This is not true if $k = \mathbb{R}$. Just consider $\frac{1}{x^2+y^2}$.

§6 Presheaf and Sheaf

§6.1 Presheaf and Sheaf

- A presheaf $\mathcal{F}$ of rings on a topological space X is a contravariant functor from the topological space X (a category with open sets as objects and inclusion as morphisms) to the category of rings. $\mathcal{F}$ sends each open set of a ring $\mathcal{F}(U)$ and each inclusion of open sets $V \subseteq U$ to a restriction ring homomorphism $\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V) : \phi_{|U} \mapsto \phi_{|V}$.
  For each open set $U \subseteq X$, the elements of $\mathcal{F}(U)$ are called the sections of $\mathcal{F}$ over U.

  We also define $\mathcal{F}(\emptyset) = 0$ (the zero ring). This is just a convention, and it makes sense because $\emptyset$ is the initial object in the category of open sets of X while 0 is the final object in the category of rings. We can simply ignore this property in most situations.

  From on now, when we are discussing about sheaf on algebraic sets, $\boxed{\text{we always assume any sheaf } \mathcal{F}}$

- A sheaf is a presheat $\mathcal{F}$ that satisfies the gluing property: If $U \subseteq X$ is an open set, for any open covering of $U = \bigcup_{i \in I} U_i$, if we have $\phi_i \in \mathcal{F}(U_i)$ such that $\phi_{i|U_i \cap U_j} = \phi_{j|U_i \cap U_j}$ for all $i, j \in I$, then there exists a unqiue lift: that is a unique $\psi \in \mathcal{F}(U)$ such that $\psi_{|U_i} = \phi_i$ for all $i \in I$.

Example 6.1 (Sheaf of Regular Functions)
Let X be an algebraic set with Zariski's topology. For any open set U, define $\mathcal{O}_X(U)$ to be the set of all regular functions on U for each open set $U \subseteq X$. Then $\mathcal{O}_X$ is a sheaf of rings on X.

Example 6.2 (Sheaf of Continuous Functions)
Let X be a subset of $\mathbb{R}^n$ with Euclidean topology. For any open set U, define $C^0(U, \mathbb{R})$ to be the set of all continuous real-valued functions on U. Then $C^0(-, \mathbb{R})$ is a sheaf of rings on X.

Example 6.3 (A presheaf of constant functions, but not a sheaf)
Let X be a subset of $\mathbb{R}$ with Euclidean topology. For any open set U, define $C^{\text{constant}}(U, \mathbb{R})$ to be the set of all constant real-valued functions on U. Then $C^{\text{constant}}(-, \mathbb{R})$ is a presheaf of rings on X. But it's not a sheaf because the gluing of constant functions may not be constant.

Remark 6.4. The reason example 3 fails to be a sheaf is that the condition is global. In general, local conditions are easier to satisfy gluing condition.

Remark 6.5. Intuitively, a presheaf on U, $\mathcal{F}(U)$ is a set of functions defined on the open set U of a topological space. This is why we say the presheaf **on** topological

> space X. And sheaf are presheaf such that the functions can be glued together nicely or define locally.

§6.2 Stalks and germs

- Let $\mathcal{F}$ be a presheaf on a topological space X. For any point $p \in X$, we define the stalk of $\mathcal{F}$ at p to be the set

$$\{(U, \phi) \mid \text{ open set } U \subseteq X, a \in U, \phi \in \mathcal{F}(U)\}/\sim$$

. $\sim$ is an equivalence relation defined as follows: $(U, \phi) \sim (V, \psi)$ if and only if there exists an open set $L \subseteq U \cap V$ such that $p \in L$ and $\phi_{|L} = \psi_{|L}$. The equivalence class of (U, ϕ) is called the germ of ϕ at p and denoted by ϕ_p. The stalk of $\mathcal{F}$ at p is denoted by $\mathcal{F}_p$.

> **Remark 6.6.** The germs in stalks are the function defined locally around the point p. So stalks capture the local behavior of the presheaf/sheaf at point p. The value at any point other than p doesn't matter (we can always find a smaller neighborhood U that doesn't contain that point).

Example 6.7

Consider the sheaf of differentiable functions on $\mathbb{R}$. The germs of a function f at point p captures the information about the value of f at p and all its derivative at p.

Example 6.8

Consider the sheaf of locally polynomial functions on $\mathbb{R}$. The stalk at point p is isomorphic to the ring of $\mathbb{R}[x]$.

Next, we will see some basic properties of stalks in the sheaf of regular functions on algebraic sets.

- We denoted the stalk of the sheaf of regular functions $\mathcal{O}_X$ at point p by $\mathcal{O}_{X,p}$. The most important information about $\mathcal{O}_{X,p}$ is that it's isomorphic to the localization of the coordinate ring $\mathcal{O}(X)$ at the maximal ideal $\mathfrak{m}_p$ corresponding to point p,

$$\{\frac{f}{g} \mid f, g \in \mathcal{O}(X), g(p) \neq 0\}$$

. And it's a local ring with a unique maximal ideal $\mathfrak{m}_{X,p} = \{\phi_p \in \mathcal{O}_{X,p} \mid \phi(p) = 0\}$. The isomorphism is given by sending $\frac{g}{f}$ in the localization $\mathcal{O}(X)_{\mathfrak{m}_p}$ to $(U_f, \frac{g}{f})$ in the stalk $\mathcal{O}_{X,p}$.

> **Remark 6.9.** An algebraic fact about local rings: A ring is a local ring if and only if the set of non-unit elements forms an ideal. In our case, the set of non-unit elements is exactly $\mathfrak{m}_{X,p}$.

§7 Ringed Spaces and its morphisms

§7.1 Ringed Space and Affine Varieties

- A ringed space is a topological space X equiped with a sheaf of rings $\mathcal{O}_X$ on X. The sheaf $\mathcal{O}_X$ is called the structure sheaf of the ringed space $(X, \mathcal{O}_X)$.
- If U is an open set of a ringed space $(X, \mathcal{O}_X)$, then $(U, \mathcal{O}_{X|U})$ is also a ringed space where $\mathcal{O}_{X|U}$ is the restriction of $\mathcal{O}_X$ on U, which is also a sheaf. ($\mathcal{O}_{X|U}(V) := \mathcal{O}_x(V)$ for all open set $V \subseteq U$)
- An affine variety is a ringed space that is **isomorphic as ringed space** to $(X, \mathcal{O}_X)$ for some algebraic set X with Zariski's topology and $\mathcal{O}_X$ is the sheaf of regular functions on X.
- Well, what is the morphism of ringed space?
- First of all, ringed spaces are topological spaces so we need the morphism to be some kind of continuous map f from X to Y.
- Next, we need f to preserve the sheaf structure or compatible with the sheaf functor.
  For any open set $U \subseteq Y$, by continuity, we know that U comes from the open set $f^{-1}(U)$. So we wish the function from $f^{-1}(U)$ to k defined by $f^*(\phi) = \phi \circ f$ to be a section in $\mathcal{O}_X(f^{-1}(U))$. In other words, we define the pullback of a section $\phi \in \mathcal{O}_Y(U)$ as a function from $f^{-1}(U)$ to k defined by $f^*(\phi) = \phi \circ f$. A morphism of ringed spaces is a continuous map from X to Y such that $f^*(\phi)$ to be a section in $\mathcal{O}_X(f^{-1}(U))$ for any open set $U \subseteq Y$.

  Then, a morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a continuous map $f : X \to Y$ such that for any open set $U \subseteq Y$ and any section $\phi \in \mathcal{O}_Y(U)$, the pullback $f^*(\phi) \in \mathcal{O}_X(f^{-1}(U))$.
- Properties of morphisms of ringed spaces:
  1. The composition of two morphisms of ringed spaces is still a morphism of ringed spaces. And the identity is also a morphism of ringed spaces. So we can form a category with ringed spaces as objects and morphisms of ringed spaces as morphisms.
  2. If f is a morphism from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$, let $U \subseteq X$ and $V \subseteq Y$ be an open sets such that $f(U) \subseteq V$, then $f_{|U}$ is a morphism of ringed spaces from $(U, \mathcal{O}_{X|U})$ to $(V, \mathcal{O}_{Y|V})$.
  3. Converse to 2, if for any open covering (U_i) of X, $f_{|U_i}$ is a morphism of ringed spaces from $(U_i, \mathcal{O}_{X|U_i})$ to $(Y, \mathcal{O}_Y)$, then f is a morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$.

     Proof. For any open set $V \subseteq Y$ and any section $\phi \in \mathcal{O}_Y(V)$,

     $$f^*(\phi)_{|U_i \cap f^{-1}(V)} = \phi \circ f_{|U_i} \in \mathcal{O}_X(U_i \cap f^{-1}(V)).$$

     By the gluing property of sheaf, we have $f^*(\phi) \in \mathcal{O}_X(f^{-1}(V))$. Therefore, f is a morphism of ringed spaces. □

- An isomorphism of ringed spaces is a morphism of ringed spaces that admits an inverse morphism of ringed spaces. (Just like homeomorphism in topology)

Example 7.1 (Bijective morphism is not isomorphism)
Consider the map from $\mathbb{A}^1$ to $V(y^2 - x^3) \subseteq \mathbb{A}^2$ defined by $t \mapsto (t^2, t^3)$. This is a bijective morphism of ringed spaces but not an isomorphism of ringed spaces because the inverse map is not a morphism of ringed spaces.

But why? How do we show it's not a morphism of ringed spaces? Motivated by this question, we need a more concrete description of morphisms of ringed spaces between affine varieties.

- The morphisms on affine varieties are the morphisms of ringed spaces between them. But we can describe them more explicitly:

Theorem 7.2
If $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$ are algebraic sets. For any open set $U \subseteq X$, if f is a morphism from $(U, \mathcal{O}_{X|U})$ to $(Y, \mathcal{O}_Y)$, then $f = (f_1, f_2, \dots, f_n)$ such that $f_1, f_2, \dots, f_n \in \mathcal{O}_{X|U}(U) = \mathcal{O}_X(U)$.

Proof. If $f : U \to Y$ is a morphism, consider $V = Y$ and $y_i \in \mathcal{O}_Y(Y)$. $f^* y_i = y_i \circ f = f_i \in \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(U)$.
Conversely, if $f = (f_1, f_2, \dots, f_n)$ such that $f_i \in \mathcal{O}_X(U)$, for any open set $V \subseteq Y$ and any section $\phi \in \mathcal{O}_Y(V)$, we have $f^*\phi = \phi \circ f = \phi(f_1, f_2, \dots, f_n) \in \mathcal{O}_X(U)$ because regular functions are closed under composition. Therefore, f is a morphism of ringed spaces. □

- With the explicit description of morphisms of ringed spaces between affine varieties, we can link the morphisms of ringed spaces to the homomorphisms of their coordinate rings:

Theorem 7.3
There is a one-to-one correspondence between the morphisms of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ and the k-algebra homomorphisms from $\mathcal{O}(Y)$ to $\mathcal{O}(X)$.

Proof. Given a morphism of ringed spaces $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, f^* is a k-algebra homomorphism from $\mathcal{O}(Y)$ to $\mathcal{O}(X)$.

Now, we construct its inverse map. Given a k-algebra homomorphism $g : \mathcal{O}(Y) \to \mathcal{O}(X)$, we define a function f from X to $\mathbb{A}^m$ (the ambient space of Y) as follows:

$$f = (g(y_1), g(y_2), \dots, g(y_m))$$

If the range of f is contained in Y, by the theorem above, f is a morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$.
Now, we show that the image of f is contained in Y. To show this, by Nullstellensatz,

we need to show $I(im(f)) \supseteq I(Y)$. For any $h \in I(Y)$, we have $g(h) = g(0) = 0$ in $\mathcal{O}(X)$. So $g(h) = h \circ f = 0$ on X. Thus, h vanishes on $im(f)$. Therefore, $I(im(f)) \supseteq I(Y)$ and $im(f) \subseteq Y$.
Finally, we can easily verify that these two maps are inverses of each other. □

> **Remark 7.4.** This theorem is very important because it allows us to study geometric morphisms using algebraic homomorphisms. Beyond that, it shows that the category of affine varieties is (contravariantly) equivalent to the category of finitely generated reduced k-algebras.
> This gives us a powerful tool to see if two affine varieties are isomorphic or not by checking if their coordinate rings are isomorphic or not.

- Now, we can come back to the example above. The map from $\mathbb{A}^1$ to $V(y^2 - x^3) \subseteq \mathbb{A}^2$ defined by $t \mapsto (t^2, t^3)$ is not an isomorphism of ringed spaces. It's clearly that it's a bijective morphism of ringed spaces. However, if it's an isomorphism, then $\mathcal{O}(V(y^2 - x^3)) \cong \frac{k[x,y]}{(y^2-x^3)}$ must be isomorphic to $\mathcal{O}(\mathbb{A}^1) = k[t]$ under the map $x \mapsto tot^2$ and $y \mapsto t^3$. But $\mathcal{O}(V(y^2 - x^3)) = k[x, y]/(y^2 - x^3)$ is not isomorphic to $k[t]$ because $k[x, y]/(y^2 - x^3)$ is not a UFD while $k[t]$ is a UFD. Therefore, the map is not an isomorphism of ringed spaces.

§7.2 Product of affine Varieties

- We wish to define the product of two affine varieties $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$. First of all, the product needs to be a topological space. So we may consider $X \times Y$ with the product topology (open sets are generated by $U \times V$ where U is open in X and V is open in Y). However, this has some problem:

> **Example 7.5**
> $V(x - y) \subseteq \mathbb{A}^2$ is closed in $\mathbb{A}^2$ with Zariski's topology. However, if we consider $\mathbb{A}^1 \times \mathbb{A}^1$ with product topology, then $V(x - y)$ is not closed. (Exercise)

- Therefore, we need to define a better topology on $X \times Y$. In fact, we can define a Zariski's topology on $X \times Y$ because $X \times Y$ is a closed subset of $\mathbb{A}^{n+m}$!

- We can verify that this definition of topology on $X \times Y$ satisfies the universal property of product in the category of ringed spaces.

$$\begin{tikzcd}
Z \arrow[dr, "f"] \arrow[drr, "f_X"] \arrow[ddr, "f_Y"'] & & \\
& X \times Y \arrow[r, "\pi_X"] \arrow[d, "\pi_Y"] & X \\
& Y &
\end{tikzcd}$$

We can apply our functor $\mathcal{O}_{\_}$ to get the following diagram in the category of k-algebras:

$$\begin{tikzcd}
\mathcal{O}(Z) & & \\
& \mathcal{O}(X \times Y) \arrow[ul, "f^*"'] & \mathcal{O}(X) \arrow[l, "\pi_X^*"'] \arrow[ull, "f_X^*"'] \\
& \mathcal{O}(Y) \arrow[u, "\pi_Y^*"] \arrow[uul, "f_Y^*"] &
\end{tikzcd}$$

So $\mathcal{O}(X \times Y)$ satisfies the universal property of tensor product in the category of k-algebras. Since the functor $\mathcal{O}_{\_}$ is an equivalence of categories between the category of affine varieties and the category of finitely generated reduced k-algebras, we have:

$$\mathcal{O}(X \times Y) \cong \mathcal{O}(X) \otimes_k \mathcal{O}(Y)$$

§7.3 Some strange affine varieties

- In fact, affine varieties don't have to be algebraic sets.

Example 7.6

Principal open set U_f of an algebraic set X can be seen as an affine variety. Let $Y := \{(x,t) \in U_f \times \mathbb{A}^1 \mid tf(x) = 1\} \subseteq \mathbb{A}^{n+1}$. Then, $(Y, \mathcal{O}_Y)$ is an affine variety, and it's isomorphic to $(U_f, \mathcal{O}_{X|U_f})$ as ringed space via the projection map $\pi : X \to U_f$. The inverse map is given by $x \mapsto (x, \frac{1}{f(x)})$.
The coordinate ring of Y is given by:

$$\mathcal{O}(Y) \cong \frac{\mathcal{O}(X)[t]}{(tf-1)} \cong \mathcal{O}(X)_f$$

Example 7.7 ($\mathbb{A}^2 \setminus \{0\}$ is not an affine Varieties)

This is clear because we have shown that the regular functions on $\mathbb{A}^2 \setminus \{0\}$ is $k[x,y]$. If it's an affine variety, then its coordinate ring must be isomorphic to $k[x,y]$. So $\mathbb{A}^2 \setminus \{0\}$ must be isomorphic to $\mathbb{A}^2$ as ringed spaces, clearly a contradiction.

§8 Prevarieties and Varieties

§8.1 Constructing Prevarieties

The problem with affine varieties is that not all "nice" spaces are affine varieties. For example, $\mathbb{A}^2 \setminus \{0\}$ is not an affine variety. So we need to generalize the notion of affine varieties to include more "nice" spaces. This leads us to the notion of prevarieties and varieties.

Definition 8.1. A prevariety is a ringed space X such that X can be covered by finitely many open sets $U_i \subseteq X$ such that each $(U_i, \mathcal{O}_{X|U_i})$ is an affine variety (that is isomorphic to some $(Y, \mathcal{O}_Y)$ for some algebraic set Y as ringed space). U_i are called affine open sets of the prevariety X. Morphisms of prevarieties are defined to be morphisms of ringed spaces.

> **Remark 8.2.** The open cover is not part of the data of prevariety. The existence of different finite open cover by affine varieties leads to the same prevarieties if prevarieties are isomorphic as ringed spaces.

> **Remark 8.3.** It's clear that any affine variety is a prevariety. Moreover, any open set of a affine variety is a prevariety because any open set of an affine variety can be covered by finitely many principal open sets, which are all affine varieties.

The basic way to construct prevarieties is to glue affine varieties together or to glue finitely many prevarieties. Let's first glue two prevarieties together $(X_1, \mathcal{O}_{X_1})$ and $(X_2, \mathcal{O}_{X_2})$. Let $U_{12} \subseteq X_1$ and $U_{21} \subseteq X_2$ be open sets such that there exists an isomorphism of ringed spaces $f_{12} : U_{12} \to U_{21}$. Therefore, we can glue X_1 and X_2 together along U_{12} and U_{21} via f_{12}. More precisely,

- As a set, we define $X = (X_1 \sqcup X_2)/\sim$ where $\sim$ is the equivalence relation defined as follows: $a \sim b$ if $a = b$ or $a \in U_{12}, b \in U_{21}$ and $f_{12}(a) = b$ or the other way around.

- As a topological space, we first define the topological space on $X_1 \sqcup X_2$ to be the coproduct of topological spaces X_1 and X_2, that is the finest topological space such that the inclusion maps $i_1' : X_1 \to X_1 \sqcup X_2$ and $i_2' : X_2 \to X_1 \sqcup X_2$ are continuous. Next, we define the topology on X to be the quotient topology induced by the quotient map $\pi : X_1 \sqcup X_2 \to X$, that is the finest topology on X such that π is continuous.

$$\begin{tikzcd}
X_1 \arrow[dr, "i_1'"] \arrow[drr, bend left, "i_1"] & & \\
& X_1 \sqcup X_2 \arrow[r, "\pi"'] & X = (X_1 \sqcup X_2)/\sim \\
X_2 \arrow[ur, "i_2'"'] \arrow[urr, bend right, "i_2"'] & &
\end{tikzcd}$$

Therefore, $U \subseteq X$ is open if and only if $i_1^{-1}(U)$ is open in X_1 and $i_2^{-1}(U)$ is open in X_2.

- As a ringed space, for any open set $U \subseteq X$, we define

$$\mathcal{O}_X(U) := \{\phi : U \to K \mid i_1^*\phi \in \mathcal{O}_{X_1}(i_1^{-1}(U)) \text{ and } i_2^*\phi \in \mathcal{O}_{X_2}(i_2^{-1}(U))\}$$

  This is indeed a presheaf of rings on X: For $V \subseteq U$ open sets of X, we have $i : V \to U$ the inclusion map. The axoims of presheaf follows from the diagram below:

$$\begin{tikzcd}
i_j^{-1}(V) \arrow[rr, "i'"] \arrow[d, "i_j"'] \arrow[ddr, bend left=10, "i_j^* i^* \phi"'] \arrow[ddr, bend left=30, "i'^* i_j^* \phi"] \arrow[drr, phantom, "\circlearrowright"] & & i_j^{-1}(U) \arrow[d, "i_j"] \arrow[ddl, "i_j^*\phi"'] \\
V \arrow[rr, "i"] \arrow[dr, bend right, "i^*\phi"'] & & U \arrow[dl, bend left, "\phi"] \\
& K &
\end{tikzcd}$$

  This is a sheaf because the sheaf axioms follow from the sheaf axioms of $\mathcal{O}_{X_1}$ and $\mathcal{O}_{X_2}$. Therefore, $(X, \mathcal{O}_X)$ is a ringed space.

One can immedaitely check the i_1 and i_2 are indeed morphisms of ringed spaces from $(X_1, \mathcal{O}_{X_1})$ and $(X_2, \mathcal{O}_{X_2})$ to $(X, \mathcal{O}_X)$ respectively. In fact, the image of $i_j(X_j)$ is an open subset of X isomorphic to $(X_j, \mathcal{O}_{X_j})$ as ringed spaces for $j = 1, 2$. We will just open the inclusion and say X_1 and X_2 are open subsets of X from now on. Since X_1 and X_2 are prevarieties, we can cover them by finitely many affine open sets. Therefore, X can be covered by finitely many affine open sets. Thus, X is a prevariety.

Example 8.4

We can glue two copies of $\mathbb{A}^1$ together via the isomorphism $f_{12} : \mathbb{A}^1 \setminus \{0\} \to \mathbb{A}^1 \setminus \{0\}$ defined by $f_{12}(x) = \frac{1}{x}$. The resulting prevariety is called the projective line and denoted by $\mathbb{P}^1$.

We can also define a map from $\mathbb{P}^1$ to $\mathbb{P}^1$ by gluing two maps $f_1 : X_1 \to X_2 \subseteq \mathbb{P}^1$ and $f_2 : X_2 \to X_1 \subseteq \mathbb{P}^1$ defined by $f_1(x) = x$ and $f_2(x) = x$. Notice that these two maps agree on the intersection $X_1 \cap X_2 = \mathbb{A}^1 \setminus \{0\}$ because for any $x \in \mathbb{A}^1 \setminus \{0\}$, $f_2(f_{12}(x)) = f_2(\frac{1}{x}) = \frac{1}{x} = f_1(x)$. Therefore, we can glue f_1 and f_2 together to get a morphism from $\mathbb{P}^1$ to $\mathbb{P}^1$. By sending x to $\frac{1}{x}$ in each copy of $\mathbb{A}^1$, we get an automorphism of $\mathbb{P}^1$ that swaps 0 and ∞.

Example 8.5

we can glue two copies of $\mathbb{A}^1$ together via the isomorphism $f_{12} : \mathbb{A}^1 \setminus \{0\} \to \mathbb{A}^1 \setminus \{0\}$ defined by $f_{12}(x) = x$. The resulting prevariety is called the affine line with double origin.

We can also get a morphism by gluing two identity maps on each copy of $\mathbb{A}^1$. The resulting morphism ϕ is an automorphism that swaps the two origins.

This is valid as a prevarieties but somewhat counterintuitive when we compare it with affine variety because $\phi(x) = x$ for any $x \neq 0$ but $\phi(0_1) = 0_2$ and $\phi(0_2) = 0_1$. So the set of fixed points of ϕ is $\mathbb{A}^1 \setminus \{0\}$, which is not closed in this prevariety. This never happens in affine varieties. We cerntainly need to avoid that.

More generally, we can glue finitely many prevarieties together to get a new prevariety. The process is similar to gluing two prevarieties together. We just need to make sure that the isomorphisms of the patches are compatible with each other: for any i, j, k and the isomorphisms $f_{ij} : U_{ij} \to U_{ji}$, $f_{jk} : U_{jk} \to U_{kj}$ and $f_{ik} : U_{ik} \to U_{ki}$. They need to satisfies

1. $f_{ij}^{-1} = f_{ji}$
2. $f_{ij}^{-1}(U_{jk}) \subseteq U_{ik}$
3. $f_{jk} \circ f_{ij} = f_{ik}$ on $f_{ij}^{-1}(U_{jk})$

The construction of ringed space structure on the glued space is similar to the case of gluing two prevarieties together. We omit the details here.

Remark 8.6. One may ask why do we need to satisfy those compatibility conditions? And how do we ensure that those conditions are suffices for us to if we want our glued space to be somewhat meaningful and well-defined?

To answer these questions, we need to recall how we construct topological spaces by gluing: taking disjoint union and quotienting by the equivalence relation $\sim$ defined as $a \sim f(a)$. The compatibility conditions ensure that the relation is indeed equivalence relation. The first condition ensures symmetry. The second and third conditions ensure transitivity. Reflexivity is clear. Therefore, the compatibility conditions are necessary and sufficient (sufficiency is not very clear if you see it in a purely geometric way) for us to construct the glued space.